

The AI committee **rejected** the cheapest, greenest mix.

Because the bricks would have failed on the factory floor. That judgment — cheaper, greener, **but never below spec** — is MixMind.

Gemma 4 12B · LoRA fine-tuned

AMD Radeon W7900 + Instinct MI300X · ROCm

strength model R^2 0.910 · measured

[open source](#)

[LIVE — hellotravisss.github.io/mixmind/web](#)

Concrete is **8% of global CO₂**. The know-how to fix a mix sits in labs most plants can't afford.

And no plant will ever upload its recipes to a third-party AI — the mix design is the business.

~8%

of global CO₂ is cement & concrete

1 in 1000s

plants has a mix-R&D department

0

recipes a plant will send to a cloud API

So the intelligence has to run **inside the fence** — on hardware the plant owns.

github.com/Hellotravisss/mixmind

A plant's R&D department, self-hosted on its own AMD GPU.

Floor Copilot

RAG · cited

Answers the plant's own shop-floor knowledge — silo bridging, curled edges, wet sand — cites the note id, and refuses when it doesn't know.

Mix Committee

4 agents · grounded

Standard, QC, Cost and Carbon officers debate a recipe change against a real strength model, and stamp a verdict.

Facts live in a **swappable knowledge base** — point the same fine-tuned model at another plant's notes and it serves that plant. The recipe never leaves the building.

Material is stuck in the cement silo – what do I do?

Give a few firm hits near the outlet with a rubber mallet to break the bridge **KB-07**. If it still won't drop, lock out the feed first, then open the inspection cover near the base **KB-07**.

What is the current market price of grey cement?

I don't have a note on that.

100%

grounded citation accuracy — 9/9 held-out phrasings, correct note cited, distractor note avoided

0

invented answers — off-notes questions get an honest refusal, by training and by design

Every claim in the debate is a computed number.

PROPOSED CHANGE	28-DAY MPA	2-DAY · DEMOULD ≥12	COST/UNIT	CO ₂	VERDICT
Sweet spot	52.6	16.1 ✓	\$87.9	-17%	APPROVED
20% cement → slag	46.3	12.7 ✓ tight	\$88.2	-18%	CONDITIONS · 1-week pilot
Aggressive · 45% slag	36.0	8.0 ✗	\$84.8	-40%	REJECTED

Strength: gradient boosting on **1,030 real lab tests** (UCI · Yeh 1998), held-out **R²=0.910, MAE 3.46 MPa**. Cost & CO₂: deterministic arithmetic. The LLM writes the debate — **never the numbers**.

REJECTED

The 45%-slag mix saves the most money and cuts CO₂ 40% — and the committee kills it.

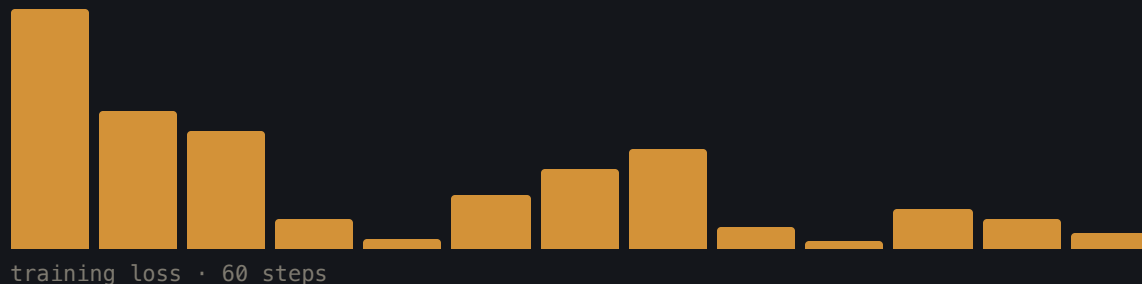
Predicted 2-day strength: **8.0 MPa**, below the 12.0 demould spec — the units would stick and chip at demould. The model learned slag's early-strength penalty **from the data**; nobody hard-coded that rule. Judgment, not a yes-man.

QC officer: "I cannot sign off on this."

github.com/Hellotravisss/mixmind

A 287 MB adapter turns Gemma 4 into **this plant's** engineer.

LoRA $r=16$ · 65.5M trainable (0.55% of 12B)
42-example plant SFT set · 3 epochs · 60 steps
loss 3.029 → 0.001 · **<1 min on MI300X**
saved adapter → mixmind-gemma4-lora (287 MB)



Measured before → after on held-out prompts:

diagnosis answers now **lead with the likely culprit** instead of paraphrasing the note

-15% length tighter, plant-voice answers

judge 2/3 independent **Fireworks LLM-judge** (Kimi K2, blind A/B) preferred the fine-tuned answer on 2 of 3 held-out prompts, tie on the third

judged **Fireworks LLM-judge** (blind A/B, independent model family): preferred the fine-tuned answer **2 of 3** held-out prompts

honest citations were already 100% with prompting — fine-tuning bakes the **behavior into the weights**, and each plant retrains on its own floor data

Everyone proves AMD can go **big**. We prove it goes **small enough to afford**.

Radeon PRO W7900 · 48 GB ~\$4k · in the plant

GPU[0] Card Vendor: Advanced Micro
Devices, Inc. [AMD/ATI] · gfx1100
VRAM Total: 51,522,830,336 B

Runs the full product — fine-tuned Gemma 4, Copilot, Committee.
Private AI a small plant can actually buy.

Instinct MI300X · 192 GB \$1.99/hr · for training

GPU[0] Card Series: AMD Instinct
MI300X VF · gfx942
VRAM Total: 205,822,885,888 B

Fine-tune in under a minute on AMD Developer Cloud, ship the 287 MB
adapter back to the plant card.

Train big, serve cheap — **both ends AMD**, verified with rocm-smi on
real hardware (evidence in repo).

The failures we hit — and shipped through.

env	!pip installs missed the Jupyter kernel on the AMD notebook — deps must go in with %pip, then restart
versions	Gemma 4 needs transformers ≥ 5.13 , which conflicts with the preinstalled ROCm vLLM (< 5) — we serve via transformers, vLLM serving is a next step
tokenizer	transformers 5.x apply_chat_template dtype crash in our trainer — fixed with return_dict=True (train_lora.py)
model	with no notes provided, base and fine-tuned model can hallucinate a citation — v2 dataset adds "no notes → no citation" examples
honesty	citation accuracy was already 100% at base with prompting — we report it as a system metric, not a fine-tune gain

We predict **directions**. We don't certify mixes.

The strength model is trained on wet-cast lab data; dry-cast paver production differs. So MixMind proposes and screens — the plant's QC still owns the final call, and load-bearing changes go to a qualified engineer. **That division of labor is the product, not a disclaimer.**

A mid-size plant spends **~\$1M+ a year** on cement. We sell the tool that trims it.

Mid-size paver/block plant (~30 t cement/day) → **~\$1.0–1.5M/yr** cement spend

SCM substitution + trimming over-design → **5–19% less cement** committee-screened, spec held

Recovered per plant → **\$50K–250K / plant-yr** + proportional CO₂ cut

MixMind at **\$2K/plant/mo**, self-hosted on one AMD GPU → **<20% of savings** pays back in the first month

The alternative is a materials engineer (**\$90–120K/yr**) or a consultant retainer — which is why most of the ~3,000 North-American precast plants simply don't optimize. That's a **\$0.3–0.8B/yr** recoverable-spend wedge, and the on-prem AMD box is what makes the price point possible.

Every plant that can't afford a research lab gets one it can run itself.

1

GPU in the plant office — the whole stack, on-prem

swap the KB

same model serves the next plant — knowledge is retrieval, not weights

tonnes

of CO₂ per plant-year from cement cut at held strength

Small precast plants, block producers, builders in underserved regions
— the operators who touch the mix every day, finally with an R&D
department that answers back. **Built by one of them.**

hellotravisss.github.io/mixmind/web

- ① **Ask the floor:** type a real shop-floor question — or try "what's the market price of cement?" and watch it **refuse** instead of guess.
- ② **The committee:** pick "**Aggressive · 45% slag**" — the cheapest, greenest mix — and watch QC stamp it **REJECTED** on early strength.
- ③ **Drive the mix:** drag cement down until the verdict flips; type **your** cement price into the price panel and watch the economics recompute.

-17%CO₂, strength held**R² 0.910**

strength model · 1,030 lab tests

100%

grounded citations · 9/9 held-out

0 bytes

of recipe data sent off-site

White cement, grey cement, slag, chips — **I touch these every day.**

MixMind is a machine operator's answer to a question the industry keeps asking: what if the plants that make the world's most-used material could afford the intelligence to make it cleaner?

Open the live demo. Ask it a floor question. Pull the cement down.